

$$\Psi(X) = \frac{1}{\sqrt{K}} (A - e^{ix} + A - e^{-ix}) \quad X < 0 \quad G_m = R_m - \frac{1}{2} R_{gm} = \frac{8\pi G}{c^3} T_{mg}$$

$$K_s = \sqrt{2\pi E/h^2} \quad S_B = \frac{k_B 4\pi G}{hc} M^2 \quad D = \frac{24\pi^3 L^2}{T^2 c^2 (1-c^2)}$$

$$H = \frac{PP}{Z_m} + V(s) \quad L = \text{tr} \left\{ \frac{1}{g} F_{12} F^{12} - i \lambda \Gamma^{\mu} \Gamma_{\mu} \lambda \right\} \quad S = \frac{1}{2k} \int R \sqrt{-g} d^4 x$$

$$R_m - \frac{1}{2} R_{gm} + \Lambda g_m = \frac{8\pi G}{c^3} T_{mg} \quad H|\Psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle \quad I = \int e^{-ix/k} dx = \sqrt{\frac{2\pi}{a}} \quad \frac{d}{dt} \langle A \rangle = \frac{1}{\hbar} \langle [\bar{A}, H] \rangle + \left\langle \frac{\partial \bar{A}}{\partial t} \right\rangle \quad S = \frac{c^3 K A}{4 h G}$$

$$Re[\Psi(x)] \quad P = -i\hbar \bar{\nabla} \quad R dF/A \quad (B+C)^2 \quad (B)^2 \quad \frac{S(k_1+k_2)}{k^2} \quad D = \frac{24\pi^3 L^2}{T^2 c^2 (1-c^2)} \quad S_f = \langle F|S|f/2 \rangle$$

$$r = \frac{\theta}{2\pi} + \frac{4\pi}{g^2} \quad \int \frac{d}{dt} \frac{2H^3}{8A} y = \frac{2H^3}{8A} \quad \vec{r}(t_0) - \vec{r}(t_1) \quad \frac{v(t_0) - v}{y = (t+b)} \quad \frac{1}{\sqrt{2}} \frac{\pi - y}{\pi - y} \quad \frac{1}{\sqrt{2}} \frac{\pi - y}{\pi - y} \quad \frac{1}{\sqrt{2}} \frac{\pi - y}{\pi - y}$$

$$S = vt + \frac{1}{2} at^2 \quad \frac{1}{\sqrt{2}} \frac{\pi - y}{\pi - y} \quad \frac{1}{\sqrt{2}} \frac{\pi - y}{\pi - y} \quad \frac{1}{\sqrt{2}} \frac{\pi - y}{\pi - y} \quad \frac{1}{\sqrt{2}} \frac{\pi - y}{\pi - y}$$



Escuela Politécnica de Enxeñaría de Ferrol

ÁLGEBRA

GRADO EN INGENIERÍA MECÁNICA

$$I = \int e^{-ix/k} dx = \sqrt{\frac{2\pi}{a}} \quad S_B = \frac{k_B 4\pi G}{hc} M^2 \quad \Psi(X) = \frac{1}{\sqrt{K}} (A - e^{ix} + A - e^{-ix}) \quad X < 0$$

$$K_s = \sqrt{2\pi E/h^2} \quad G_m = R_m - \frac{1}{2} R_{gm} = \frac{8\pi G}{c^3} T_{mg} \quad \int \text{rot} F d\Sigma = F d\mu$$

TEMA 2. SISTEMAS DE ECUACIONES LINEALES

Factorización LU

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

Factorización LU

Matriz invertible!

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 2 & 1 \\ -3 & -2 & 2 \end{bmatrix}$$

Factorización LU

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -3/2 & 1 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 2 & 2 & -1 \\ 0 & 1 & 3/2 \\ 0 & 0 & -1 \end{bmatrix}$$

Factorización de Cholesky

Matriz simétrica y definida positiva!

$$A = \begin{bmatrix} -1 & -1 & 1 \\ -1 & 5 & -5 \\ 1 & -5 & 6 \end{bmatrix}$$

Factorización de Cholesky

Matriz simétrica y definida positiva!

$$A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 5 & -5 \\ 1 & -5 & 6 \end{bmatrix}$$

Factorización de Cholesky

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 2 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

$$L^t = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

Factorización LDU

$$A = \begin{bmatrix} -3 & 2 & -1 \\ 1 & 0 & 3 \\ 3 & 1 & 6 \end{bmatrix}$$

Factorización LDU

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ -1 & 9/2 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} -3 & 2 & -1 \\ 0 & 2/3 & 8/3 \\ 0 & 0 & -7 \end{bmatrix}$$

Factorización LDU

$$L_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ -1 & 9/2 & 1 \end{bmatrix} \quad U_1 = \begin{bmatrix} 1 & -2/3 & 1/3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 2/3 & 0 \\ 0 & 0 & -7 \end{bmatrix}$$

Discusión, rango y determinantes

$$\begin{cases} ax + y + z = 1 \\ x + ay + z = a \\ x + y + az = a^2 \end{cases}$$

Discusión, rango y determinantes

$a \neq 1, -2$	$\text{rg}(A) = 3$	$\text{rg}(A B) = 3$	SCD
$a = 1$	$\text{rg}(A) = 1$	$\text{rg}(A B) = 1$	SCI
$a = -2$	$\text{rg}(A) = 2$	$\text{rg}(A B) = 3$	SI

Regla de Cramer

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n = b_2 \\ \cdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n = b_n \end{cases}$$

Regla de Cramer

$$A \times X = B$$

$$x_1 = \frac{1}{|A|} \begin{vmatrix} b_1 & \dots & a_n \\ \dots & \ddots & \vdots \\ b_n & \dots & a_{nn} \end{vmatrix}$$

$$x_n = \frac{1}{|A|} \times \begin{vmatrix} a_{11} & \dots & b_1 \\ \dots & \ddots & \vdots \\ a_{n1} & \dots & b_n \end{vmatrix}$$

Regla de Cramer

$$\begin{cases} x - 3y + 5z = -24 \\ 2x - y + 4z = -8 \\ x + y = 9 \end{cases}$$

Regla de Cramer

$$\begin{cases} x = 7 \\ y = 2 \\ z = -5 \end{cases}$$